Af Education Services

UCE MATHEMATICS ITEM BANK

2026 Edition

SAMPLE EXTRACT

Scenario-based items with final answers and decisions

Paper 456/1

Numbers · Patterns and Algebra
Data and Probability · Geometry and Measures

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About this book

These items are grouped under the four key themes:

1. **Numbers**
2. **Patterns and Algebra**
3. **Data and Probability**
4. **Geometry and Measures**

Your Daily Practice Challenge: Complete at least 4 items every day, two from a theme you enjoy and other two from one that challenges you. ***The final answers and decisions for each theme are given in the Responses section at the end of that theme.***

Paper format (456/1)

FORMAT 456/1. Section A comprises of two compulsory items and Section B comprises of **Part I and Part II** each having two questions and a learner answers one question from a part.

Item	**Area of construct**	**Topics covered**
Item one	**Numbers** Learners appreciate and use computational skills to solve problems in real-life situations.	1. Number bases 2. Working with integers. 3. Fractions, percentages, and decimals. 4. Numerical concepts 1 and 2. 5. Ratios and proportions

Item	Area of construct	Topics covered
Item Two	**Patterns and algebra** Learners appreciate and use analysis to solve problems in real-life situations.	1. Sequences and patterns. 2. Equations of lines and curves. 3. Algebra 1 and 2. 4. Mapping and relations. 5. Inequalities and regions. 6. Equation of a straight line. 7. Rectangular Cartesian plane. 8. Simultaneous equations. 9. Linear programming. 10. Loci
Items 3 and 4	**Data and Probability.** Learner appreciates and uses logical reasoning to solve problems in real-life situations	1. Data collection and presentation. 2. Graphs. 3. Set theory. 4. Data collection and display. 5. Matrices. 6. Probability

Item	Area of construct	Topics covered
Items 5 and 6	**Geometry and measures.** Learner appreciates and uses spatial reasoning to solve real-life situations	1. Geometric construction skills. 2. Bearings. 3. General and angle properties of geometric figures. 4. Reflection. 5. Business arithmetic. 6. Time and timetables. 7. Similarities and enlargement. 8. Circles. 9. Rotation. 10. Length and area properties of two-dimensional geometrical figures. 11. Nets, areas, and volumes of solids. 12. Trigonometry 1 and 2. 13. Vectors. 14. Business mathematics. 15. Matrix transformation. 16. Circle properties. 17. Lines and planes in three dimensions

Marking and major scoring areas

Numbers

Theme	Score	Justification
Numbers	I – Identification	1. Converting from one base to another. 2. Use of percentages in real life 3. Use of ratios 4. Writing from words to figures 5. Use of directed numbers 6. Use of LCM or HCF 7. Use of BODMAS 8. Converting mixed to improper 9. Use of the formula
	M – Manipulation	1. Correct addition, subtraction, multiplication and division of values 2. Correct application of rules of indices 3. Correct use of square roots in surds 4. Approximations
	AP1 – Application	1. The ability to derive meaningful conclusions 2. If a learner derives a logical conclusion from the correct manipulation of the task

Patterns and Algebra

Theme	Score	Justification
Patterns and Algebra	F – Formation	1. Building linear equations 2. Building quadratic or simultaneous equations 3. Building inequalities 4. Building an expression
	M – Manipulation	1. Changing inequalities into lines 2. Finding points to plot 3. Solving quadratic equations 4. Plotting inequalities or straight lines, showing feasible regions 5. Reading and writing coordinates from the graph 6. Substituting for algebraic expressions
	AP2 – Application	1. Evaluating the objective function 2. Stating the number of items or unit cost of an item after solving equations

Data and Probability

Theme	Score	Justification
Data and Probability	P – Presentation	1. Writing the title of a graph 2. Writing data in columns 3. Building matrices 4. Correct use of scales 5. Labelling the graph 6. Plotting and joining correctly

Theme	Score	Justification
	A – Analysis	1. Matrix multiplication, addition and subtraction 2. Computing cumulative frequency, mid class values, class boundaries 3. Substituting in the formula of median, mode or mean 4. Determining the median position 5. Determining the mode 6. Estimating the median or mode using a graph 7. Computes probability
	IN – Interpretation	1. A learner makes a conclusion to the task

Geometry and Measures

Theme	Score	Justification
Geometry and Measures	A – Analysis	1. Arranging the given information 2. Choosing an appropriate scale 3. Sketching 4. Applying operations on given information
	M – Manipulation	1. Measuring correct angles 2. Showing compass points in bearing 3. Measuring distances 4. Converting units of length, volume, area 5. Computing tax bands 6. Computing the volume, area and distances 7. Correct use of trigonometric ratios

Theme	**Score**	**Justification**
	AP4 – Application	1. Response to task required 2. A learner concludes logically

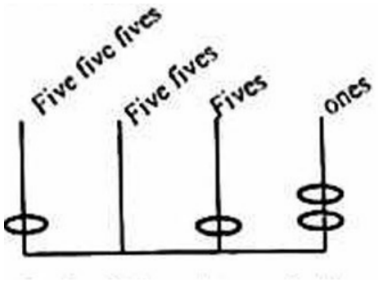
THEME 1

NUMBERS

Item 1.

Awor wants to start a business of selling shoes in her S.4 vacation. She has six hundred sixty thousand shillings and, according to her budget, she still needs two thirds of the total amount she wants. Her father has agreed to give her the remaining amount. She is not sure of how much her father is to give her.

She orders for the shoes, and the shoes are delivered in a container. She buys the container of shoes at a cost equivalent to the total amount she wants. She will sell the shoes in pairs and, on receiving the container that has the shoes, the total number of shoes is represented as shown below.



She wants to know the number of pairs of shoes that are in the container. Her target is to make a total profit of 60% but she does not know how much she must sell each pair to reach this target.

Tasks:

a) How much will Awor get from her father?

b) How many pairs of shoes are in the container?

c) Calculate the price at which Awor should sell each pair of shoes to reach her target.

Item 5.

Your guardian has a budget of Ugx 700,000 for your school expenses. To get to the school where your guardian wishes to take you for A-level, your guardian drove 4 km east from your home to the stage and then 8 km north to reach there. However, you realised later that there was a direct route from home to school your guardian could have used.

On reaching the school, you found out that the school fees, admission fees and uniform fees are Ugx 900,000, Ugx 100,000 and Ugx 350,000 respectively. The school also offers a bursary of 60% off school fees, free admission and eighty-seven thousand five hundred shillings off uniform fees to those who got first grade, and according to your results, you qualify for this bursary.

It also has two payment plans on the total amount (school fees, admission and uniform) that guardians can choose from, and they are:

- Paying in two instalments, that is to say: two thirds of the total amount at the beginning of the term and the balance at either visitation day or end of term.
- Paying in three equal instalments: at the beginning of the term, on visitation day and at the end of term respectively.

Tasks:

a) How far is it from your home to school if you travel through the direct route?

b) (i) Since you qualify for the bursary, how much will you pay?

(ii) Will your guardian afford the school expenses according to his budget?

c) (i) How much will those who don't qualify for the bursary pay per instalment, according to each of the payment plans?

(ii) Which payment plan would you recommend for them and why?

Item 12.

A man intends to plant trees on the two sides of the road which leads to his land. On one side of the road, he is to plant a tree every 5 m, yet on the other side he is to plant a tree every 6 m. At the start of the road, two trees are to be planted directly opposite each other. In the first phase, he will plant trees until another pair of trees is again directly opposite.

His land has an area of 500 m^2. He plans to use 25% of the land to plant maize, one fifth of the land for beans and 205 m^2 for growing groundnuts.

Tasks:

a) Help the man determine how many tree seedlings he needs to buy to just plant this first phase.

b) Determine the size of the land to be used for growing maize.

c) Determine the size of the land to be used for growing beans.

d) Express the area to be used for growing groundnuts in standard form.

e) Do you think he partitioned the entire land properly? Give a reason.

THEME 1: NUMBERS

RESPONSES

Item 1.

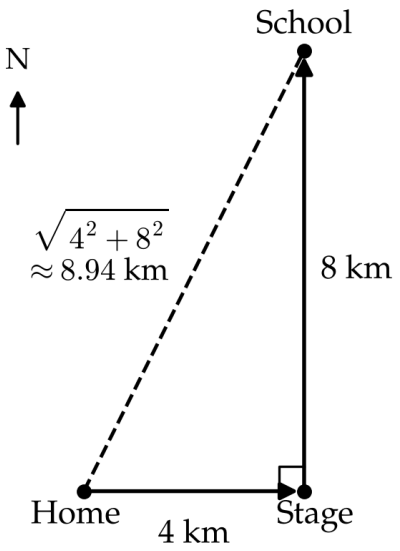
a) Ugx 1,320,000 (Ugx 660,000 is one third of the Ugx 1,980,000 she needs).

b) $1012_{five} = 132$ shoes, which is **66 pairs**.

c) Target sales Ugx 3,168,000, so she should sell each pair at **Ugx 48,000**.

Item 5.

a) $\sqrt{4^2 + 8^2} = \sqrt{80} = 4\sqrt{5} \approx$ 8.94 km.



b) (i) **Ugx 622,500**.

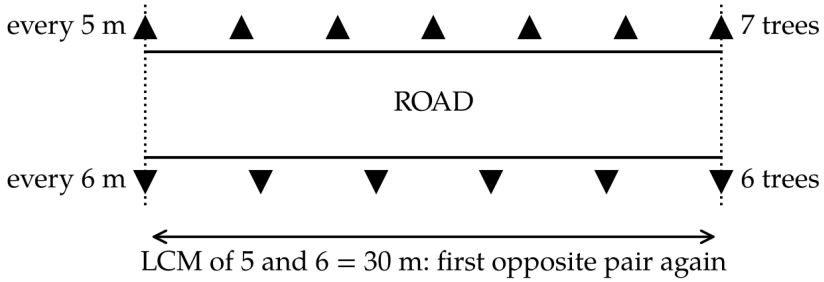
(ii) **Yes**; it is within the budget, leaving Ugx 77,500.

c) (i) Total Ugx 1,350,000. Plan 1: **Ugx 900,000** then **Ugx 450,000**. Plan 2: **Ugx 450,000** per instalment.

(ii) **Plan 2**: the first payment (Ugx 450,000) is half that of Plan 1, so the cost is spread more evenly.

Item 12.

a) LCM of 5 and 6 is 30 m: 7 + 6 = **13 seedlings**.



b) **125 m^2**.

c) **100 m^2**.

d) 2.05×10^2 m^2.

e) **No**: 70 m^2 of the land is left unassigned.

THEME 2

PATTERNS AND ALGEBRA

Item 1.

Julius and his son are to go to a certain hospital for medication. Julius wants to know whether the 4 litres of fuel he filled in his car will be enough to take them there and bring them back. His vehicle covers 7 km on 1.5 litres of fuel and 11 km on 2.5 litres of fuel, and the distance it covers is linearly related to the fuel used on each leg of the journey. The distance from the set-off point to the hospital is 17 km.

When at the hospital, Julius realises that he forgot his son's birth certificate, yet he is not sure of his son's age, and it is highly needed by the doctor. Julius tells the doctor that he remembers correctly that his age is twice the square of his son's age, and the difference between his age and his son's age is 45 years. The doctor wants to know the age of Julius' son using the remembered information.

Julius is also going to buy 4 packs of tablets and 3 bottles of syrup, but he does not know their total cost. The doctor tells him that 2 packs of tablets and a bottle of syrup cost a total of Ugx 14,500, and a pack of tablets and 2 bottles of syrup cost a total of Ugx 14,000.

Tasks:

a) (i) Determine the relationship between distance and litres of fuel for Julius' vehicle.

(ii) Use the relationship to decide whether the fuel he filled in his car will be enough. If not, how many litres of fuel will he have to add?

b) Use the information that Julius remembers to help the doctor know the age of the son.

c) How much money will Julius pay for the tablets and syrups he is to buy?

Item 2.

A farmer in Kisoko Village, Tororo District, is planning to cultivate 500 m^2 of land to plant Irish potatoes and tomatoes. In the first season last year, he hired 3 women from the neighbouring village to clear 100 m^2, while in the second season he hired 5 women to clear 300 m^2. The land cleared is linearly related to the number of women hired. He doesn't know how many women he needs this season to clear the intended land size.

This season he is planting Irish potatoes and tomatoes at the same time. He is planting at most 3 boxes of Irish potatoes. He has been told that the total number of boxes of seedlings of both Irish potatoes and tomatoes must not be less than 6, and because of space, the number of boxes of tomatoes must be less than 6. Each box of Irish potatoes yields Ugx 180,000, while each box of tomatoes yields Ugx 200,000. He wants to obtain the maximum yield.

Tasks:

a) (i) Formulate a mathematical relationship between the number of women and the size of land cleared.

(ii) How many women will he need to clear the land this season?

b) (i) Write down mathematical inequalities to represent the above conditions.

(ii) Show clearly the feasible region on a Cartesian plane.

(iii) How many boxes of Irish potatoes and tomatoes should the farmer plant in order to obtain the maximum yield?

Item 5.

Your family doesn't have enough money to organise a big graduation party for your elder brother Paul after graduating as a civil engineer. Paul suggests that they use Fare Hotel near the university, where they will invite a few family members and friends to a simple graduation party.

A plate of food for an adult is Ugx 20,000 and for a child is Ugx 10,000. The space agreed upon is for 50 people. The family has Ugx 900,000 to cater for the food at this hotel. The chairperson of the organising committee doesn't know whether to register 40 adults with 10 children or 35 adults with 15 children.

Some family members and friends missed space at Fare Hotel; among them were both family members and friends. They were identified as follows: at least 12 adults in total, and the family members (x) were at least three times the number of friends (y). The family solicited some money to cater for those who missed space, and the contributions amounted to Ugx 130,000. Those who had missed were taken to a nearby restaurant, where every family member was given food at Ugx 10,000 and every friend at Ugx 15,000.

Tasks:

a) Help the chairperson to decide on the number of adults and children to register in order to use all the planned money.

b) Using a graph, determine the number of family members and friends whose bills at the restaurant the solicited money can clear without leaving any money.

THEME 2: PATTERNS AND ALGEBRA

RESPONSES

Item 1.

a) (i) $y = 4x + 1$, where y is the distance in km and x is the fuel in litres.

(ii) **No**: 17 km needs $x = \dfrac{17-1}{4} = 4$ litres, so the 4 litres only take them to the hospital. He must **add 4 litres** for the return leg.

b) $2s^2 - s - 45 = 0$ gives $s = 5$: the son is **5 years old** (Julius is 50).

c) Tablets Ugx 5,000 per pack, syrup Ugx 4,500 per bottle: **Ugx 33,500**.

Item 2.

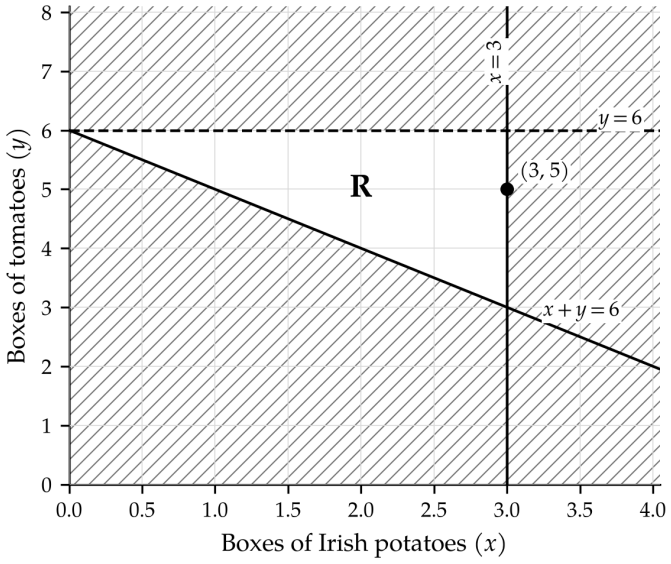
a) (i) $y = 100x - 200$ (x women, y m^2 cleared).

(ii) **7 women**.

b) (i) With x boxes of Irish potatoes and y boxes of tomatoes: $x \leq 3,\ x + y \geq 6,\ y < 6,\ x \geq 0,\ y \geq 0$.

(ii) The region bounded by $x = 3$, $x + y = 6$ and $y = 5$ (whole-number points only).

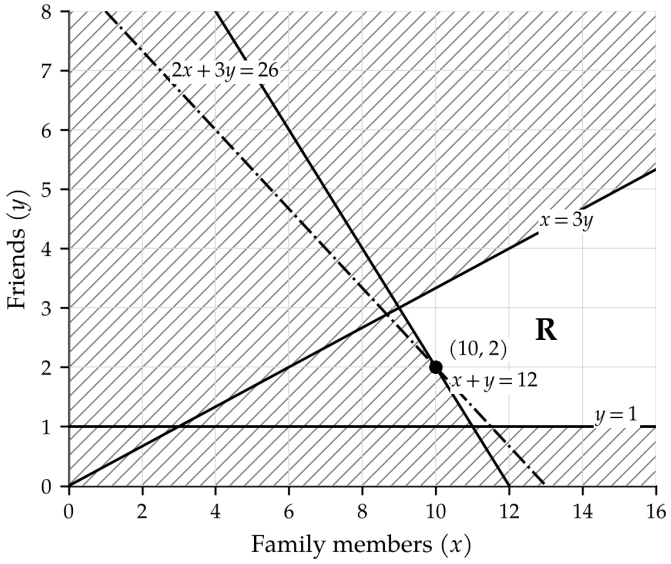
(iii) **3 boxes of Irish potatoes and 5 boxes of tomatoes**, giving the maximum yield of **Ugx 1,540,000**.



Item 5.

a) $a + c = 50$ and $2a + c = 90$: **40 adults and 10 children**.

b) $10,000x + 15,000y = 130,000$ with $x + y \geq 12,\ \ x \geq 3y,\ \ y \geq 1$: **10 family members and 2 friends**.



THEME 3

DATA AND PROBABILITY

Item 1.

The Ministry of Health, through Nagongera Health Centre IV, has embarked on efforts to reduce tuberculosis (TB). The director of the health centre has tasked the laboratory technician to provide information on the ages of all patients who tested positive for TB, to find out which ages are likely to recover from TB.

The ages of the patients that visited the health centre last month are shown below.

57	12	71	83	66	25	51	33	62	54
17	54	19	73	45	70	62	51	94	38
23	44	43	20	40	72	36	43	37	28
64	25	59	61	42	37	41	65	89	57
55	58	45	72	69	78	59	35	57	23

The doctors noted that 60% of the patients recover from TB, and that it is the youngest 60% of patients who usually recover.

Tasks:

a) What is the average age of the patients that visited the health centre last month?

b) Using a suitable graph, determine the ages of the patients who are likely to recover from TB.

Item 8.

36 farmers of Mwero Zone A formed an association, and in the recent harvest they got the following weights of produce (in kg).

240	250	360	400	240	160	250	320	210	330	350	380
270	290	390	300	340	150	240	310	200	230	290	280
170	240	340	310	180	150	250	280	290	280	250	360

All the farmers agreed to bring together all their produce and share it equally, to support those who got low produce to remain in production. It is assumed that every farmer will receive 270 kg after sharing equally.

The Ministry of Agriculture, Animal Industry and Fisheries would like to support not more than ten farmers, from those whose produce did not exceed 210.5 kg, with farm inputs. The fertiliser factory in Tororo is going to facilitate farmers who got more than 310 kg of produce to visit the factory, to educate them about the effective use of artificial fertilisers.

Tasks:

a) Organise the above information so that it is well understood, and state whether the assumption was right.

b) With the help of a statistical illustration, determine whether the support was to be given or not, and if it was given, how many farmers got it.

c) What is the chance that a member of the association is going to visit the fertiliser factory in Tororo?

Item 15.

A company holds daily morning briefings at 8:05 a.m. for all its workers, who walk to work from different places of residence. The supervisor notices that some workers miss important communication because they arrive late. He decided to collect data about their arrival times so that he can make the necessary adjustments and also help some of them to become residents near the company premises. The arrival times, in minutes after 8:05 a.m., were as follows.

52	49	11	16	32	28	32	55	38	24
15	23	33	20	40	39	22	37	53	30
29	58	38	44	31	21	18	39	25	47
51	49	35	24	47	34	48	44	55	19
27	38	41	59	33	52	27	36	46	46
32	28	47	54	34	21	58	37	26	42

Tasks:

a) With a reason based on calculations from the data above, suggest the appropriate time at which the morning briefings should always start.

b) The company manager suggests that the first 25% of the workers to arrive be given a transport refund, those between 25% and 75% be given accommodation, and the rest be stopped. Determine the latest arrival time for one to qualify for;

   (i) a transport refund.

   (ii) accommodation.

c) How many workers would be stopped from work?

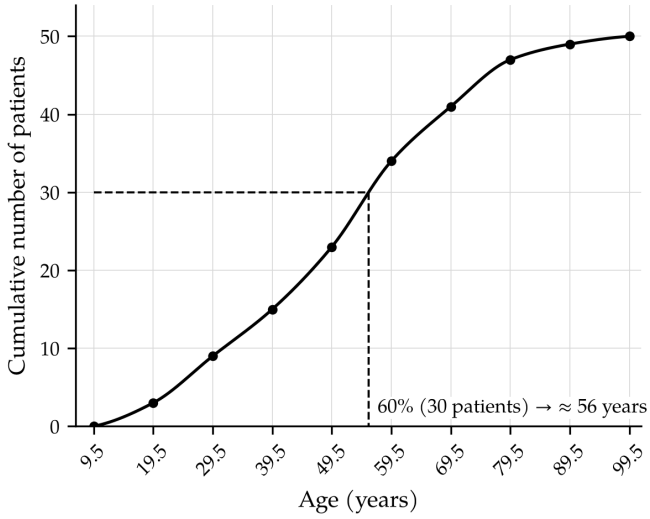
THEME 3: DATA AND PROBABILITY

RESPONSES

Item 1.

a) $\dfrac{2514}{50} =$ **50.3 years** (grouped mean with classes 10–19, 20–29, ...: 50.3 years).

b) From the ogive, 60% of the patients (30 patients) are below about **56 years**, so patients **younger than about 56 years** are likely to recover.



Item 8.

a) In a frequency table (classes 150–199, 200–249, ...), the total is 9,880 kg, so each farmer gets $\dfrac{9880}{36} \approx 274.4$ kg. **The assumption of 270 kg was not right**: each farmer gets about 274 kg.

b) From the ogive (or the table), 7 farmers had at most 210.5 kg. Since $7 \leq 10$, **the support was given to 7 farmers**.



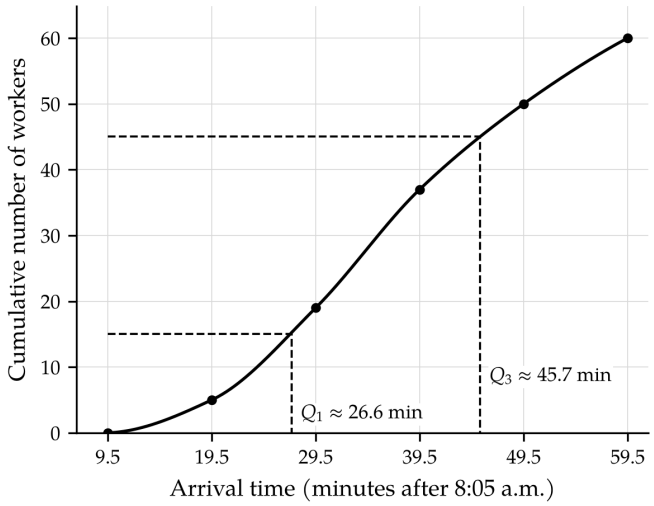
c) 10 farmers had more than 310 kg: $\frac{10}{36} = \frac{5}{18}$

Item 15.

a) Mean arrival 36.5 min after 8:05 a.m., so start at **about 8:42 a.m.**

b) (i) Lower quartile ≈ 26.6 min: by **about 8:32 a.m.**

(ii) Upper quartile ≈ 45.7 min: by **about 8:51 a.m.**



c) 25% of 60 = **15 workers**.

THEME 4

GEOMETRY AND MEASURES

Item 5.

James is responsible for transporting goods from a distribution centre to three retail stores. His delivery van has a fuel tank that holds 25 litres of fuel when full, and it consumes 1 litre for every 12 km travelled. He starts with a full tank.

James begins his trip from the distribution centre to the first retail store, which is 84 km away on a bearing of N55°E. After completing the delivery, he continues due east at an average speed of 50 km/h for 30 minutes to the second retail store. From there, he changes direction, takes a bearing of 210°, and drives a distance of 60 km to the third retail store.

At the third retail store, James is informed of a direct route back to the distribution centre, but he does not know the exact distance. He is uncertain whether the remaining fuel in his van will be sufficient for the journey.

Tasks:

Help James determine:

a) the distance of the direct route from the third retail store to the distribution centre;

b) whether the remaining fuel in the van is sufficient for the journey using the direct route.

Item 13.

Your aunt has started a baking business. She wants to make a new cake base in the shape of a regular pentagon with sides of 4 cm. The cake base will be cut from a circular piece of cardboard, so she asks you to draw the cake base accurately for her.

The first customer to order the new design asked for 10 cakes, each priced at Ugx 150,000. The production cost per cake is Ugx 100,000, and she has promised the client a discount of 2%. She is also required to pay a sales tax of Ugx 1,000 for every Ugx 10,000 she earns from her sales. After paying the tax, she deposits the rest of the money in the bank. She needs your help to determine how much she will deposit in the bank when the client pays.

Tasks:

a) Show how you would accurately draw the cake base.

b) Show your aunt how to determine the amount of money that she will deposit in her bank account.

Item 23.

Your elder brother Moses, who is 1.5 m tall, stands on top of your uncle's 34 m building to take pictures of an event. From the top, he sees a car and a bicycle on the straight road below, on the same side of the building. Using his phone's angle feature, he records the angle of depression of the car as 50° and of the bicycle as 65°. He challenges you to calculate how far apart the car and the bicycle are.

During the day, a magician entertains the village with an Easter special. One of his tricks involves a bag containing red and white balls. He says the bag has x red balls and $(x-8)$ white balls, and that the probability of drawing a red ball is $\frac{2}{3}$. He promises a gift to anyone who can find the total number of balls in the bag.

Tasks:

a) Calculate how far the bicycle is from the car.

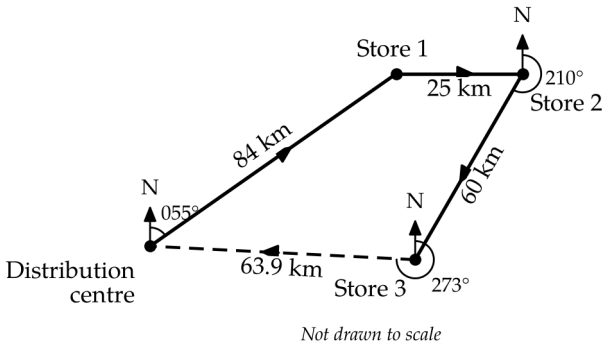
b) Find the total number of balls in the magician's bag.

THEME 4: GEOMETRY AND MEASURES

RESPONSES

Item 5.

a) **63.9 km**.

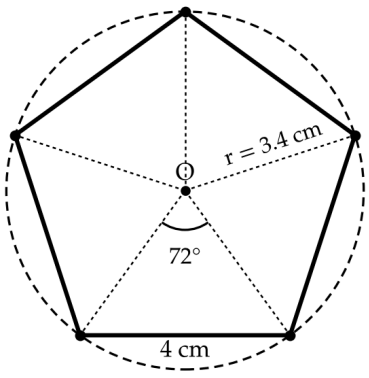


Not drawn to scale

b) Fuel used so far $= \dfrac{84+25+60}{12} = 14.1$ L, so 10.9 L remain. The direct route needs $\dfrac{63.9}{12} \approx 5.3$ L. **Yes, the fuel is sufficient.**

Item 13.

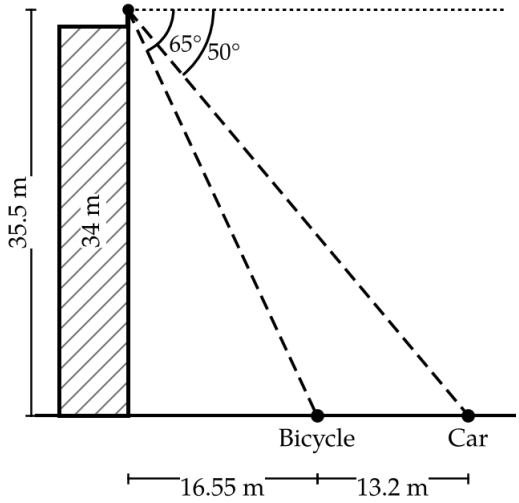
a) Each side subtends $\dfrac{360^\circ}{5} = 72^\circ$ at the centre, so the radius of the circle is $\dfrac{2}{\sin 36^\circ} =$ **3.4 cm**. Draw a circle of radius 3.4 cm, step off five chords of 4 cm (or five 72° angles at the centre) and join the points.



b) Sales $= 10 \times 150,000 \times 0.98 =$ Ugx 1,470,000; tax (10%) = Ugx 147,000. **Deposit = Ugx 1,323,000** (her profit after the Ugx 1,000,000 production cost is Ugx 323,000).

Item 23.

a) Eye level = 35.5 m. $\dfrac{35.5}{\tan 50^\circ} - \dfrac{35.5}{\tan 65^\circ} = 29.79 - 16.55 = \mathbf{13.2\ m}$.



b) $\dfrac{x}{2x-8} = \dfrac{2}{3}$ gives $x = 16$, so there are **24 balls** (16 red and 8 white).

Other books by the same author

1. S.1 Mathematics Workbook
2. S.2 Mathematics Workbook
3. S.3 Mathematics Workbook
4. S.4 Mathematics Workbook
5. NLSC Item Bank
6. **UACE MATHEMATICS ITEM BANK, PAPER 1**

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For God and My country